

SOFTWARE ARCHITECTURE 101

MODERN TECH ARCHITECTURE IS BRO

Why 80% of engineers struggle with production
implementation—and the exact blueprint to fix it.

THE STANDARD

REST API

The most widely used API type. Uses standard HTTP methods (GET, POST, PUT, DELETE).

- ✓ Lightweight and easy to use
- ✓ Returns data mostly in **JSON format**
- ✓ Stateless architecture

BEST FOR

Web & Mobile Apps,
Public APIs

EXAMPLES

Social Media,
Weather, E-commerce

THE VAULT

SOAP API

A protocol for highly secure and structured communication.

- ✓ Uses **XML format** over HTTP/HTTPS
- ✓ Highly secure (built-in ACID compliance)
- ✓ Works with strict WS-* standards

BEST FOR

Enterprise Systems,
Banking

EXAMPLES

Payment Gateways,
Govt Services

— THE SURGEON

GraphQL **API**

Zero Over- fetching.

A query language for APIs that lets you fetch exactly the data you need.



- ✓ Single endpoint **for all requests**
- ✓ Reduces payload size
- ✓ Faster data retrieval

1 URL

replaces dozens of REST
endpoints.

☐ **Modern Web Apps**

☐ **Mobile Applications**

— REAL-TIME & PERFORMANCE

Streaming & **Microservices**

SERVER

📠 **WebSocket API**

Two-way communication enabling low latency and instant real-time updates. The connection stays open.

USE CASES: CHAT APPS, LIVE GAMING

⚡ **gRPC API**

High-performance framework by Google. Very fast, uses **Protocol Buffers instead of JSON.**

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